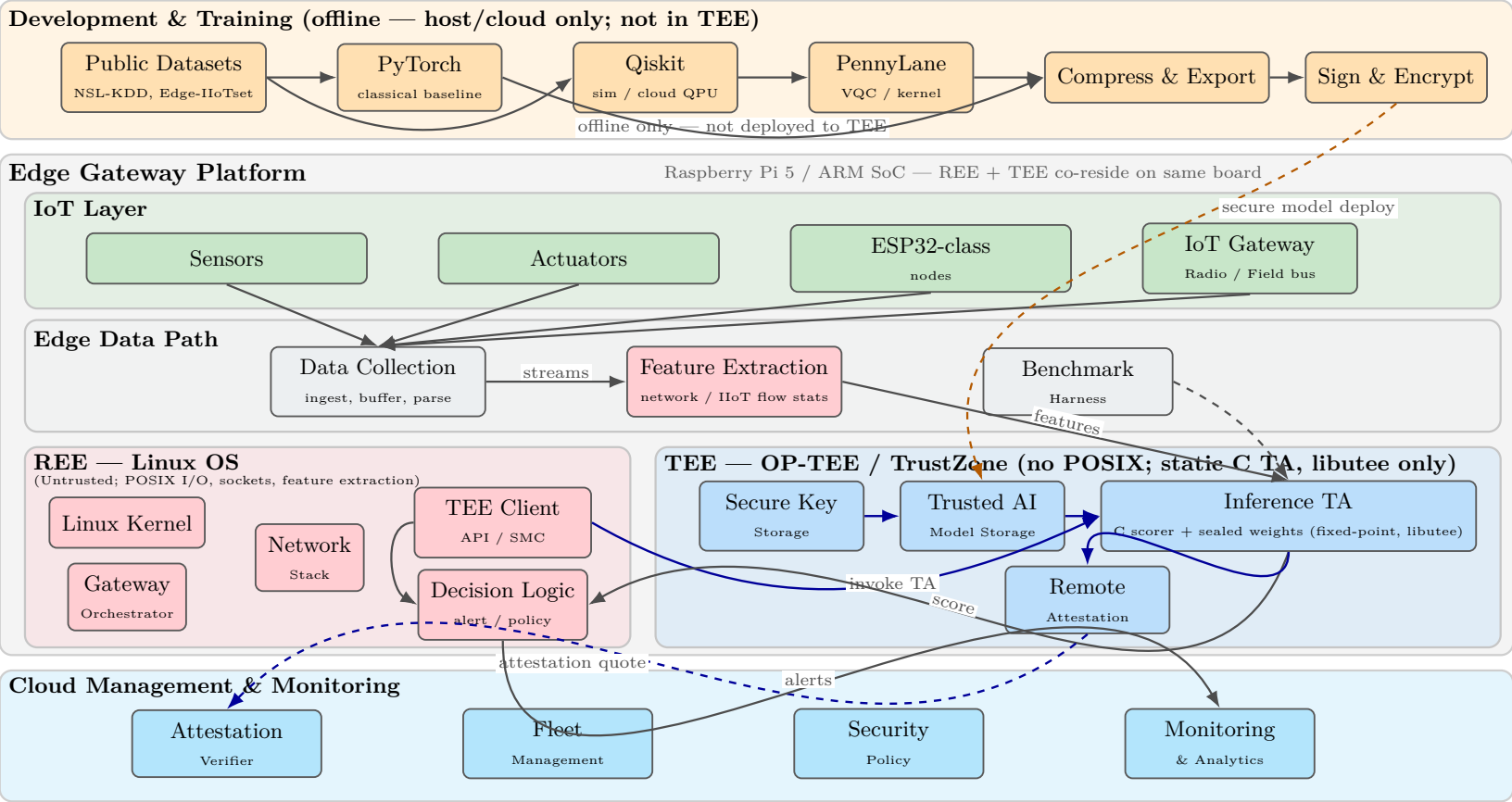


Trustworthy Hybrid Quantum-Classical AI for Secure IoT Edge Systems

Reference architecture (TRL 2-3) — Raspberry Pi 5 / ARM gateway; REE handles I/O, OP-TEE TA runs protected C inference only



Flow and methodology:

- *Phase A (offline)*: train PyTorch TinyML baseline and hybrid models (Qiskit/PennyLane on simulator/cloud); compress to a **quantum-inspired C surrogate**; sign and encrypt; deploy to trusted model storage.
- *Phase B (online, REE)*: ingest network/IIoT telemetry (flow stats, protocol fields, gateway metadata) — **not** primary host HIDS (syscalls/file access); extract feature vector; invoke TA via libtee.
- *Phase B (online, TEE)*: Inference TA loads sealed weights and runs **fixed-point C scoring only** (no Qiskit, no POSIX, no sockets); returns anomaly score; attestation quote optional.
- *Phase C (cloud)*: verify attestation, fleet policy, monitoring/alerts. REE decision logic may block/log based on score — QAI is **not an Agent AI**.

Implementation assumptions (Raspberry Pi 5 + OP-TEE; targets to be measured):

- *Not Agent AI*: trusted **anomaly scorer** inside a TA — no autonomous planning, tool use, or IPS enforcement inside the secure world.
- *Input data*: NSL-KDD / Edge-IIoTset / Bot-IoT network & IIoT telemetry features (packet/flow statistics); feature extraction in REE; TA receives a fixed-size float/int vector via TEE_InvokeCommand.
- *Resource budgets*: REE feature extract $\leq 5-10$ ms; TA inference $\leq 10-50$ ms; end-to-end $T \leq 100$ ms; TA heap **256 KB–1 MB**, stack 32–64 KB, model weights 50–200 KB compressed (TA_DATA_SIZE in manifest).
- *OP-TEE feasibility*: TA uses libutee only; all network/file I/O stays in REE client; Qiskit/PennyLane never run inside TrustZone. Prototype milestone: passthrough TA → classical C inference TA → hybrid surrogate TA.
- *Prototype status*: architecture defined (TRL 2-3); Raspberry Pi 5 + OP-TEE passthrough TA and classical inference TA are **in progress** — measured results to follow; full hybrid surrogate TA is a Year 2 deliverable.

Research questions:

RQ1. Why is TinyML insufficient; why consider Quantum AI?

- TinyML = compressed *classical* boundaries from known attacks; recall drops on **stealthy / distribution-shift** IIoT traffic (Edge-IIoTset DDoS/DoS/recon variants).
- Under fixed P , T , and model size, aggressive quantization limits capacity; hybrid Q-C surrogates are a **testable alternative** benchmarked vs. PyTorch→TFLite TinyML baseline.

RQ2. Roles of Quantum AI and TEE

- *QAI*: anomaly **detection** engine (scoring function); **not** a security primitive; hypothesis — better recall@FPR on unseen IIoT attacks at equal deployability; null results valid.
- *TEE*: hardware trust boundary — protects keys, model storage, inference runtime, attestation; mitigates tampering, unauthorized replacement/extraction/inference, REE compromise. **QAI = detection; TEE = protection.**

Primary test case (QML vs. TinyML): stealthy/low-rate IIoT gateway intrusions in Edge-IIoTset where equally sized TinyML autoencoder/CNN shows higher false negatives; compare recall@FPR, latency, joules/inference, TA memory.

— data — trusted — deploy